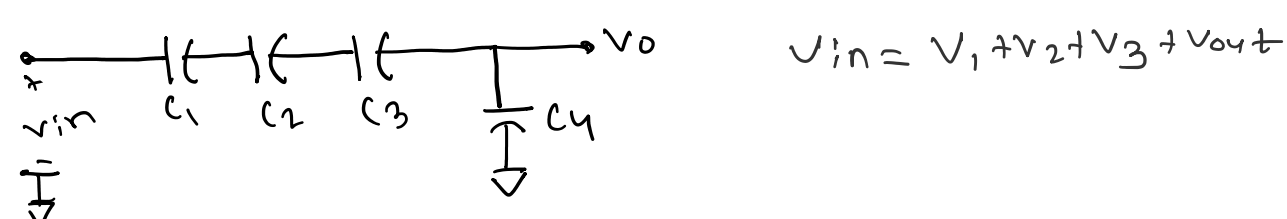
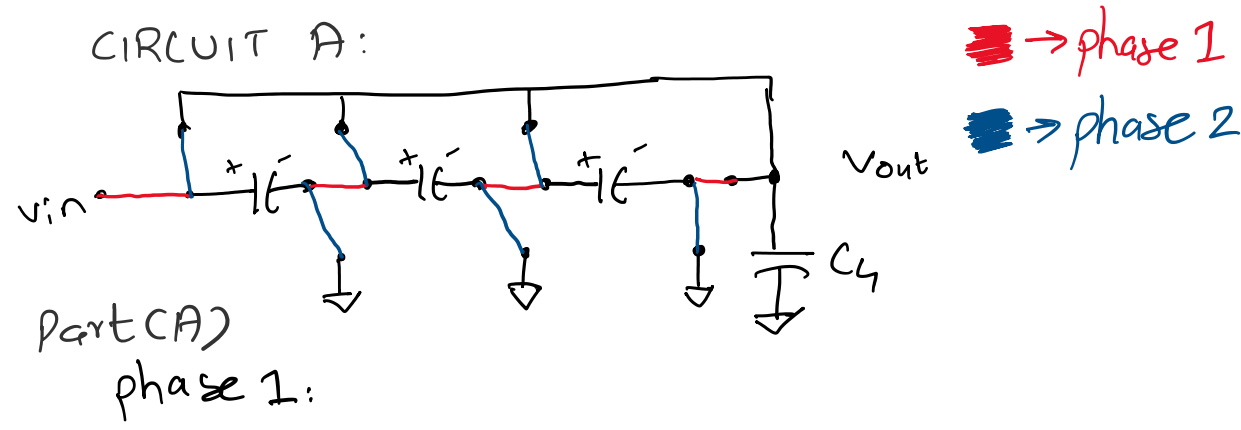
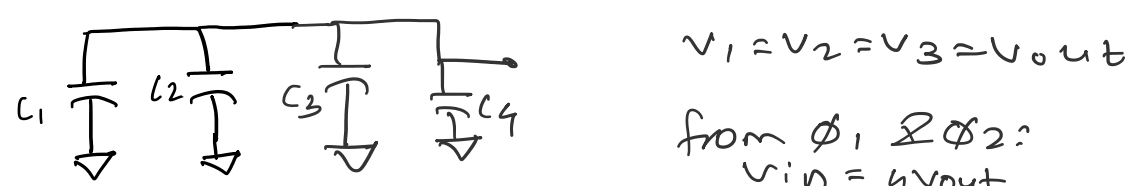


CIRCUIT A:



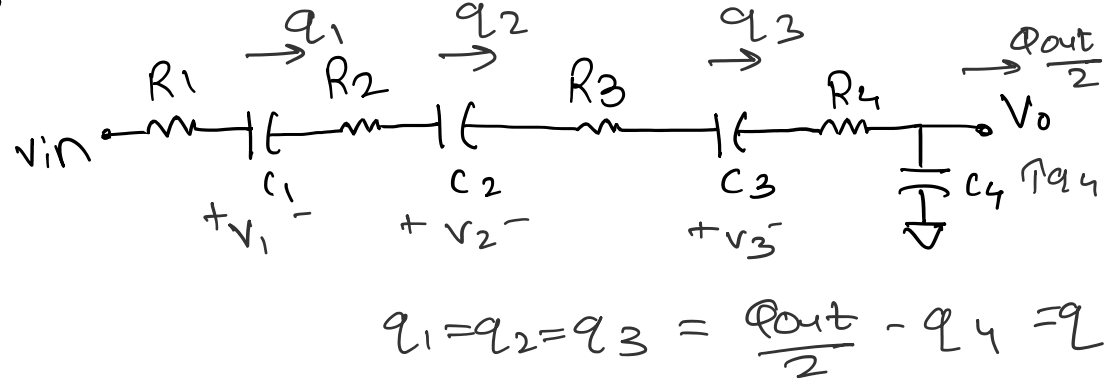
phase 2:



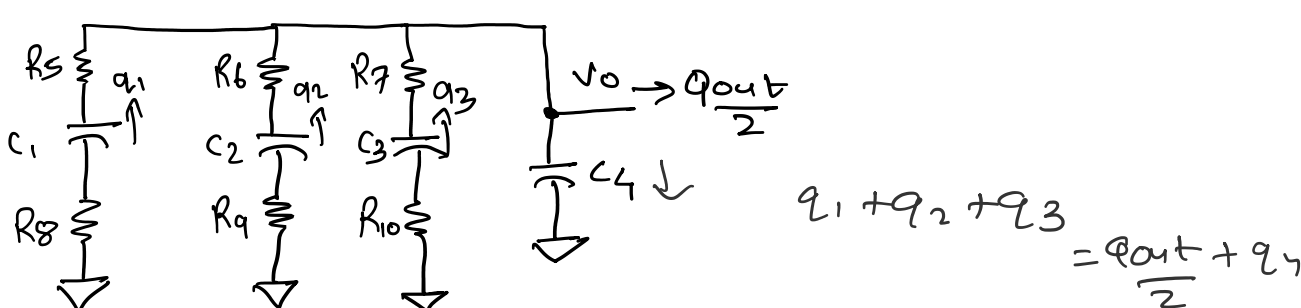
$$\frac{V_o}{V_{in}} = M = \frac{1}{4}$$

free space

Part (B)

Rout $\rightarrow$ Doing Q calculation first: ϕ_1 :

phase 2:



$$q + q_4 = \frac{Q_{out}}{2}$$

$$3q - q_4 = \frac{Q_{out}}{2}$$

$$4q = Q_{out}$$

$$q = \frac{Q_{out}}{4} = q_1 = q_2 = q_3$$

$$q_4 = \frac{Q_{out}}{2} - q \Rightarrow \boxed{q_4 = \frac{Q_{out}}{4}}$$

$$E_T = \sum_{i=1}^{10} R_i \frac{q_i^2}{\tau_i} \quad (\tau_i = \tau)$$

$$\Rightarrow E_T = \frac{2}{\tau} \times \left[\frac{q_1^2 (R_1 + \dots + R_4)}{\phi_1} + \frac{q_2^2 (R_5 + \dots + R_{10})}{\phi_2} \right]$$

$$= \frac{2}{\tau} \times \frac{Q_{out}^2}{16} \times (R_1 + \dots + R_{10})$$

$$\boxed{E_T = \frac{Q_{out}^2}{8\tau} (R_1 + \dots + R_{10})}$$

$$Q_{out} = I_{out} \tau$$

$$P = \frac{E_T}{\tau}$$

$$P = \frac{Q_{out}^2}{8\tau^2} (R_1 + \dots + R_{10})$$

$$P = \frac{I_{out}^2 \tau^2}{8\tau^2} (R_1 + \dots + R_{10})$$

$$R_{FSL} I_{out}^2 = \frac{1}{8} (R_1 + \dots + R_{10}) \times I_{out}^2$$

$$\Rightarrow \boxed{R_{FSL} = \frac{1}{8} (R_1 + \dots + R_{10})}$$

$$\text{If } R_1 = R_2 = \dots = R_{10} = R_{on}$$

$$\boxed{R_{FSL} = \frac{5}{4} R_{on}}$$

Part (C)

$$R_{FSL} = 2 \sum_{i=1}^{10} R_i \alpha_i^2$$

$$\text{so, } 2\alpha_i^2 = \frac{1}{8} \Rightarrow (\alpha_i)^2 = \sqrt{\frac{1}{16}} = \frac{1}{4}$$

$$\Rightarrow \alpha_i = [\frac{1}{4}, \frac{1}{4}, \dots, \frac{1}{4}] \quad (10 \text{ elements})$$

$$R_{FSL} = P R_{on} \Rightarrow \frac{5}{4} R_{on} = P R_{on}$$

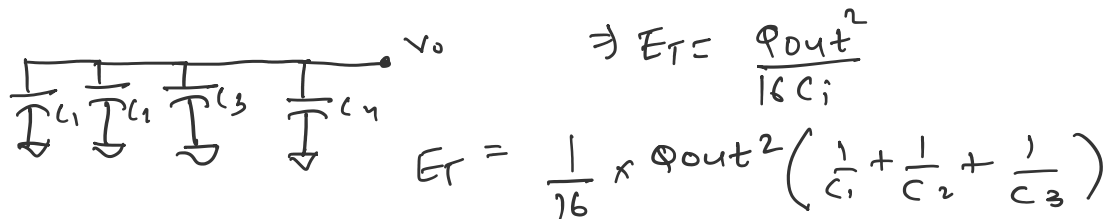
$$\Rightarrow \boxed{P = \frac{5}{4}}$$

Part (D):

Now /

SSL:

$$E_T = \sum_{i=1}^3 \frac{q_i^2}{C_i} \quad E_T = \frac{q_1^2}{C_1} + \frac{q_2^2}{C_2} + \frac{q_3^2}{C_3}$$

 ϕ_1 : ϕ_2 :

$$Q_{out} = I_{out} \tau$$

$$\Rightarrow \frac{1}{16} (I_{out}^2 \tau^2) \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right) = E_T$$

$$P = \frac{E_T}{\tau} = \frac{1}{16} \tau \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right) I_{out}^2 = R_{SSL} \cdot I_{out}^2$$

$$\Rightarrow \boxed{R_{SSL} = \frac{1}{16} \times \frac{1}{f_s} \times \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)}$$

$$\text{If, } C_1 = C_2 = C_3 = C,$$

$$R_{SSL} = \frac{1}{16} \times \frac{1}{f_s} \times \frac{3}{C} = \boxed{\frac{3}{16 C \cdot f_s} = R_{SSL}}$$

$$R_{SSL} = \frac{m}{f_s \cdot C} = \frac{3}{16 \cdot C \cdot f_s}$$

$$\Rightarrow \boxed{m = \frac{3}{16}}$$

Part (E)

 $C_T \rightarrow$ given

$$R_{out} = \sqrt{R_{SSL}^2 + R_{FSL}^2}$$

Minimum at $f = f_c$

$$\Rightarrow R_{SSL} = R_{FSL}$$

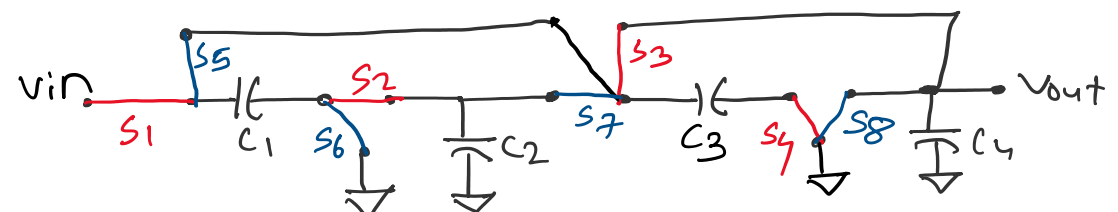
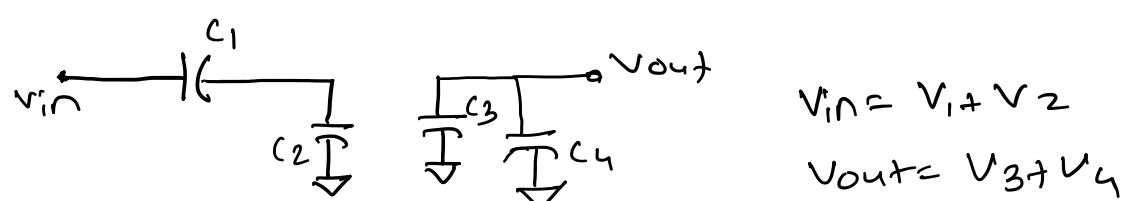
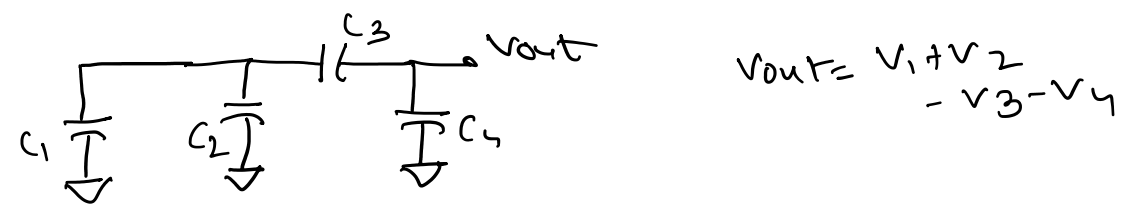
$$\Rightarrow \frac{m}{f_s C_T} = P R_{on}$$

$$\boxed{m = f_s C_T P R_{on}}$$

$$\text{for } C_T = 4.5 \text{ nF, } f_s = 5 \text{ MHz, } P = \frac{5}{4}, R_{on} = 5 \Omega$$

$$\boxed{m = 0.14}$$

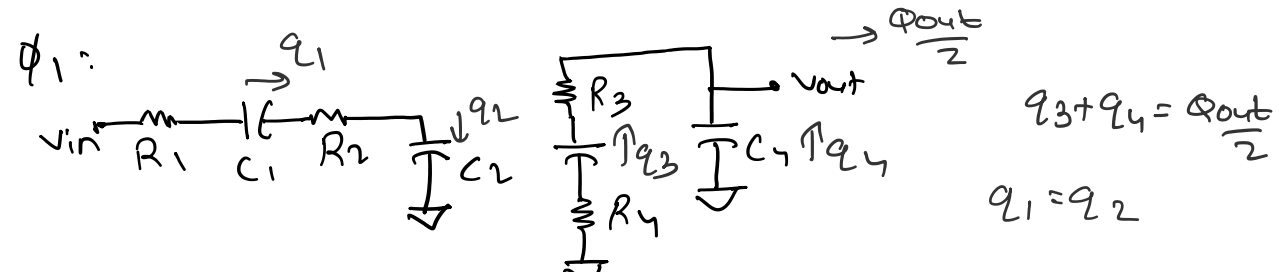
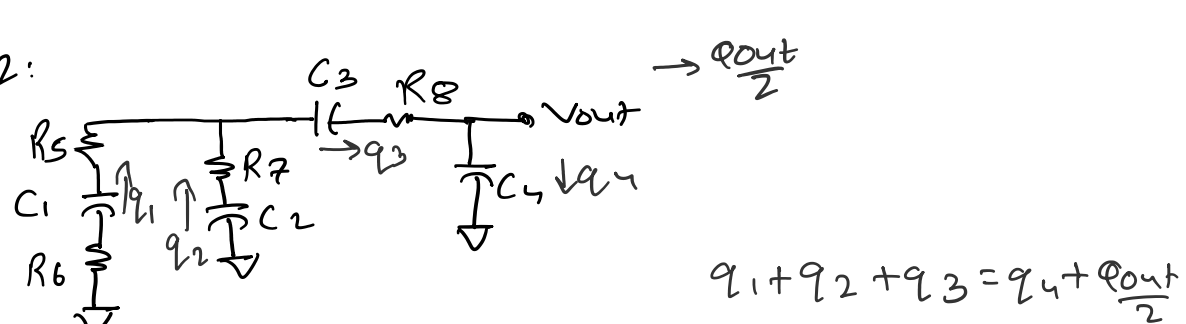
CIRCUIT B:

 ϕ_1 : ϕ_2 :

$$\Rightarrow V_{out} = V_{in} - V_{out}$$

$$2V_{out} = V_{in}$$

$$\frac{V_{out}}{V_{in}} = \boxed{m = \frac{1}{2}}$$

Rout calc: $\rightarrow$  ϕ_2 :

$$q_1 + q_2 = 2q_4$$

$$q_1 = q_2 = q_4$$

$$\Rightarrow q_3 + q_4 = \frac{Q_{out}}{2}$$

$$q_3 = q_4$$

$$\boxed{q_1 = q_2 = q_3 = q_4 = \frac{Q_{out}}{4}}$$

$$E_T = \sum_{i=1}^8 R_i \frac{q_i^2}{\tau_i} = \frac{Q_{out}^2}{16} \times \frac{2}{\tau} (R_1 + \dots + R_8)$$

$$E_T = \frac{Q_{out}^2}{8\tau} (R_1 + \dots + R_8)$$

$$= \frac{I_{out}^2 \tau}{8} (R_1 + \dots + R_8)$$

$$P = \frac{E_T}{\tau} = \frac{I_{out}^2}{8} (R_1 + \dots + R_8) \Rightarrow \boxed{R_{FSL} = \frac{R_1 + \dots + R_8}{8}}$$

$$\text{If: } R_1 = R_2 = \dots = R_8, \boxed{R_{FSL} = R_{on}}$$

$$R_{FSL} = P R_{on} \Rightarrow \boxed{P = 1}$$

Now, R_{SSL} calc:

$$E_T = \sum_{i=1}^3 \frac{q_i^2}{C_i} = \frac{Q_{out}^2}{16} \times \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

$$= \frac{I_{out}^2 \tau^2}{16} \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

$$P = I_{out}^2 \frac{1}{16 f_s} \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

$$\boxed{R_{SSL} = \frac{1}{16 f_s} \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)}$$

(same case as before)

$$\boxed{m = \frac{3}{16}} \text{ at } C_1 = C_2 = C_3$$

Circuit B provides lesser min o/p resistance,

$$[R_{FSL} = R_{min}]$$

$$\text{Ckt A: } R_{FSL} = \frac{5}{4} R_{on}$$

$$\text{Ckt B: } R_{FSL} = R_{on}$$

 R_{SSL} is same for both